DEVELOPER API REFERENCE

SAAPT 2.0 Backend Action-Based REST API Reference

Complete Documentation: Classes, Sections, Academic Years, Blood Groups, Students & Auth

BASE URL (LOCAL)
http://127.0.0.1:8000 / http://127.0.0.1:8001

API STYLE
Explicit Action Endpoints + Standard REST Conventions

AUTHENTICATION
JSON Web Token (Authorization: Bearer <token>)

DATABASE
PostgreSQL (Supabase) / SQLite Development

SUPPORTED PAYLOAD FORMATS
application/json & multipart/form-data

RELEASE VERSION
Version 2.0 • Updated October 2026

Documented Functional Modules (42 Endpoints)

- ✓ **Module 1:** Student Management CRUD (9 Endpoints)
- ✓ **Module 2:** Class & Section Allocation (6 Endpoints)
- ✓ **Module 3:** School Class Master CRUD (6 Endpoints)
- ✓ **Module 4:** Section Master CRUD (6 Endpoints)
- ✓ **Module 5:** Academic Year Master CRUD (6 Endpoints)
- ✓ **Module 6:** Blood Group Master CRUD (6 Endpoints)
- ✓ **Module 7:** Authentication & Security (4 Endpoints)

Quick Endpoint Directory

All Action URLs

Module	Operation	Method	Action Endpoint URL	Standard Router Equivalent
School Class	List / Lookup Classes	GET	`/api/list-classes/` or `/api/get-classes/`	`/api/classes/`
	Create New Class	POST	`/api/add-class/`	`/api/classes/`
	View Class Details	GET	`/api/view-class/<id>/`	`/api/classes/<id>/`
	Edit / Update Class	PUT PATCH	`/api/edit-class/<id>/`	`/api/classes/<id>/`
	Delete Class	DELETE	`/api/delete-class/<id>/`	`/api/classes/<id>/`
Section Master	List / Lookup Sections	GET	`/api/list-sections/` or `/api/get-sections/`	`/api/sections/`
	Create Section	POST	`/api/add-section/`	`/api/sections/`
	View Section Details	GET	`/api/view-section/<id>/`	`/api/sections/<id>/`
	Edit / Update Section	PUT PATCH	`/api/edit-section/<id>/`	`/api/sections/<id>/`
	Delete Section	DELETE	`/api/delete-section/<id>/`	`/api/sections/<id>/`
Academic Year	List / Lookup Academic Years	GET	`/api/list-academic-years/` or `/api/get-academic-years/`	`/api/academic-years/`
	Create Academic Year	POST	`/api/add-academic-year/`	`/api/academic-years/`
	View Academic Year Details	GET	`/api/view-academic-year/<id>/`	`/api/academic-years/<id>/`
	Edit / Update Academic Year	PUT PATCH	`/api/edit-academic-year/<id>/`	`/api/academic-years/<id>/`
	Delete Academic Year	DELETE	`/api/delete-academic-year/<id>/`	`/api/academic-years/<id>/`
Blood Group	List / Lookup Blood Groups	GET	`/api/list-blood-groups/` or `/api/get-blood-groups/`	`/api/blood-groups/`
	Create Blood Group	POST	`/api/add-blood-group/`	`/api/blood-groups/`
	View Blood Group Details	GET	`/api/view-blood-group/<id>/`	`/api/blood-groups/<id>/`
	Edit / Update Blood Group	PUT PATCH	`/api/edit-blood-group/<id>/`	`/api/blood-groups/<id>/`
	Delete Blood Group	DELETE	`/api/delete-blood-group/<id>/`	`/api/blood-groups/<id>/`

Module	Operation	Method	Action Endpoint URL	Standard Router Equivalent
Student Management	Add Student (All Steps 1-3)	POST	`/api/add-student/`	`/api/students/`
	List Students (Filter, Search, Page)	GET	`/api/list-students/`	`/api/students/`
	View Student Complete Profile	GET	`/api/view-student/<id>/`	`/api/students/<id>/`
	Edit Student	PUT PATCH	`/api/edit-student/<id>/`	`/api/students/<id>/`
	Delete Student	DELETE	`/api/delete-student/<id>/`	`/api/students/<id>/`
	Student Dashboard Statistics	GET	`/api/student-stats/`	`/api/students/stats/`

1. School Class Master CRUD APIs

Base: /api/

The School Class module manages standard academic grades (e.g. "Class 6", "Class 7", "Class 8"). Classes can be referenced by integer ID or string name when creating student profiles and section allocations.

GET /api/list-classes/ or /api/get-classes/ or /api/classes/

Bearer Token Required application/json

Retrieves all registered school classes ordered by numerical sequence. Ideal for populating UI dropdown selectors and master data tables.

RESPONSE (200 OK)

```
[
  {
    "id": 1,
    "name": "Class 6",
    "code": "CLS-6",
    "numeric_order": 6
  },
  {
    "id": 2,
    "name": "Class 7",
    "code": "CLS-7",
    "numeric_order": 7
  },
  {
    "id": 3,
    "name": "Class 8",
    "code": "CLS-8",
    "numeric_order": 8
  }
]
```

POST `/api/add-class/` or `/api/classes/` Bearer Token Required application/json

Creates a new class entry. `name` is required and must be unique. `code` and `numeric_order` are optional.

REQUEST BODY PARAMETERS

Field	Type	Required	Description
`name`	`string`	**Required**	Class display name, e.g. `"Class 11"` or `"Grade 11"` (Max 50 chars, Unique).
`code`	`string`	Optional	Unique shortcode, e.g. `"CLS-11"` (Max 20 chars).
`numeric_order`	`integer`	Optional	Sorting rank, e.g. 11 (Default: 0).

REQUEST BODY EXAMPLE

```
{
  "name": "Class 11",
  "code": "CLS-11",
  "numeric_order": 11
}
```

RESPONSE (201 CREATED)

```
{
  "id": 6,
  "name": "Class 11",
  "code": "CLS-11",
  "numeric_order": 11
}
```

GET PUT PATCH DELETE **/api/view-class/<id>/ | /api/edit-class/<id>/ | /api/delete-class/<id>/**

Bearer Token Required

Retrieve: `GET /api/view-class/6/` returns the class object (200 OK).
Update: `PUT /api/edit-class/6/` or `PATCH /api/edit-class/6/` modifies fields (200 OK).
Delete: `DELETE /api/delete-class/6/` permanently deletes the class record (204 No Content).

EDIT REQUEST BODY (PATCH EXAMPLE)

```
{
  "name": "Class 11 - Senior"
}
```

EDIT RESPONSE (200 OK)

```
{
  "id": 6,
  "name": "Class 11 - Senior",
  "code": "CLS-11",
  "numeric_order": 11
}
```

DELETE RESPONSE

```
204 No Content (Empty Response Body)
```

2. Section Master CRUD APIs

Base: /api/

The Section module manages classroom section letters (e.g. "A", "B", "C", "D"). Sections can be referenced by integer ID or string character in student forms and class-section allocations.

GET /api/list-sections/ or /api/get-sections/ or /api/sections/

Bearer Token Required application/json

Retrieves all section letters. Used to populate Section dropdowns across the application.

RESPONSE (200 OK)

```
[
  { "id": 1, "name": "A" },
  { "id": 2, "name": "B" },
  { "id": 3, "name": "C" },
  { "id": 4, "name": "D" }
]
```

POST /api/add-section/ or /api/sections/

Bearer Token Required application/json

Creates a new section. name is required and must be unique (e.g. "E").

REQUEST BODY PARAMETERS

Field	Type	Required	Description
name	string	Required	Section identifier, e.g. "E" or "Blue" (Max 10 chars, Unique).

REQUEST BODY EXAMPLE

```
{
  "name": "E"
}
```

RESPONSE (201 CREATED)

```
{
  "id": 5,
  "name": "E"
}
```

GET PUT PATCH DELETE **/api/view-section/<id>/ | /api/edit-section/<id>/ | /api/delete-section/<id>/**

Bearer Token Required

Retrieve: `GET /api/view-section/5/` returns section details (200 OK).
Update: `PUT /api/edit-section/5/` or `PATCH /api/edit-section/5/` modifies section name (200 OK).
Delete: `DELETE /api/delete-section/5/` removes the section (204 No Content).

EDIT REQUEST BODY

```
{
  "name": "Rose"
}
```

EDIT RESPONSE (200 OK)

```
{
  "id": 5,
  "name": "Rose"
}
```

DELETE RESPONSE

```
204 No Content (Empty Response Body)
```

3. Academic Year Master CRUD APIs

Base: /api/

The Academic Year module defines school terms/sessions (e.g. "2025-2026", "2026-2027"). Supports marking the active current year and date boundary validation. Can be specified by integer ID or string name in student endpoints.

GET /api/list-academic-years/ or /api/get-academic-years/ or /api/academic-years/

Bearer Token Required application/json

Retrieves all academic years. Identifies which session is flagged as the current active school year.

RESPONSE (200 OK)

```
[
  {
    "id": 1,
    "year_name": "2025-2026",
    "start_date": "2025-06-01",
    "end_date": "2026-04-30",
    "is_current": false
  },
  {
    "id": 2,
    "year_name": "2026-2027",
    "start_date": "2026-06-01",
    "end_date": "2027-04-30",
    "is_current": true
  }
]
```

POST `/api/add-academic-year/` or `/api/academic-years/` Bearer Token Required application/json

Creates a new academic year session. Setting `is_current` to `true` marks this as the default active year.

REQUEST BODY PARAMETERS

Field	Type	Required	Description
`year_name`	`string`	**Required**	Unique name, e.g. `"2027-2028"` (Max 20 chars, Unique).
`start_date`	`date (YYYY-MM-DD)`	Optional	Opening date of academic session.
`end_date`	`date (YYYY-MM-DD)`	Optional	Closing date of academic session.
`is_current`	`boolean`	Optional	Flag indicating if this is the active school year (Default: false).

REQUEST BODY EXAMPLE

```
{
  "year_name": "2027-2028",
  "start_date": "2027-06-01",
  "end_date": "2028-04-30",
  "is_current": false
}
```

RESPONSE (201 CREATED)

```
{
  "id": 3,
  "year_name": "2027-2028",
  "start_date": "2027-06-01",
  "end_date": "2028-04-30",
  "is_current": false
}
```

GET PUT PATCH DELETE **/api/view-academic-year/<id>/ | /api/edit-academic-year/<id>/ | /api/delete-academic-year/<id>/**

Bearer Token Required

Retrieve: `GET /api/view-academic-year/2/` returns the session details (200 OK).
Update: `PUT /api/edit-academic-year/2/` or `PATCH /api/edit-academic-year/2/` updates dates or flags (200 OK).
Delete: `DELETE /api/delete-academic-year/3/` removes an unused academic year (204 No Content).

EDIT REQUEST BODY (PATCH EXAMPLE)

```
{
  "is_current": true
}
```

EDIT RESPONSE (200 OK)

```
{
  "id": 2,
  "year_name": "2026-2027",
  "start_date": "2026-06-01",
  "end_date": "2027-04-30",
  "is_current": true
}
```

DELETE RESPONSE

```
204 No Content (Empty Response Body)
```

4. Blood Group Master CRUD APIs

Base: /api/

The Blood Group module manages medical blood typing groups (e.g. "A+", "A-", "B+", "B-", "O+", "O-", "AB+", "AB-", "Unknown"). Supports full CRUD and populates emergency contact & student medical registries.

GET `/api/list-blood-groups/` or `/api/get-blood-groups/` or `/api/blood-groups/`

Bearer Token Required | application/json

Retrieves all registered blood groups with medical codes and descriptions for UI dropdown selection.

RESPONSE (200 OK)


```
[
  {
    "id": 1,
    "name": "A+",
    "code": "A_POS",
    "description": "A Positive",
    "created_at": "2026-10-21T06:40:00Z"
  },
  {
    "id": 5,
    "name": "O+",
    "code": "O_POS",
    "description": "O Positive",
    "created_at": "2026-10-21T06:40:00Z"
  },
  {
    "id": 6,
    "name": "O-",
    "code": "O_NEG",
    "description": "O Negative (Universal Donor)",
    "created_at": "2026-10-21T06:40:00Z"
  }
]
```

POST /api/add-blood-group/ or /api/blood-groups/

Bearer Token Required | application/json

Creates a new blood group entry. name is required and unique.

REQUEST BODY PARAMETERS

Field	Type	Required	Description
name	string	Required	Blood group name, e.g. "Bombay Blood Group" or "Rh-null" (Max 20 chars, Unique).
code	string	Optional	Medical short identifier, e.g. "HH_POS" (Max 10 chars).
description	string	Optional	Clinical notes or donor details (Max 100 chars).

REQUEST BODY EXAMPLE

```
{
  "name": "Bombay Blood Group",
  "code": "HH_POS",
  "description": "Rare Bombay phenotype"
}
```

RESPONSE (201 CREATED)

```
{
  "id": 10,
  "name": "Bombay Blood Group",
  "code": "HH_POS",
  "description": "Rare Bombay phenotype",
  "created_at": "2026-10-21T07:15:00Z"
}
```

GET PUT PATCH DELETE **/api/view-blood-group/<id>/ | /api/edit-blood-group/<id>/ | /api/delete-blood-group/<id>/**

Bearer Token Required

Retrieve: `GET /api/view-blood-group/5/` returns blood group object (200 OK).
Update: `PUT /api/edit-blood-group/5/` or `PATCH /api/edit-blood-group/5/` updates description/code (200 OK).
Delete: `DELETE /api/delete-blood-group/10/` removes the blood group (204 No Content).

EDIT REQUEST BODY (PATCH EXAMPLE)

```
{
  "description": "O Positive - Universal RBC Recipient A/B/O"
}
```

EDIT RESPONSE (200 OK)

```
{
  "id": 5,
  "name": "O+",
  "code": "O_POS",
  "description": "O Positive - Universal RBC Recipient A/B/O",
  "created_at": "2026-10-21T06:40:00Z"
}
```

DELETE RESPONSE

```
204 No Content (Empty Response Body)
```

5. Student Management (Action-Based APIs)

Base: /api/

POST /api/add-student/ or /api/students/

Bearer Token Required | multipart/form-data & application/json

Add Student API: Creates a complete student admission profile. Supports both string names and integer IDs for `class`, `section`, and `academic_year`. Automatically creates parent/guardian records and returns all 3 steps' fields in the response.

REQUEST PARAMETERS (FORM DATA / JSON)

Field	Type	Required	Description / Example
`full_name`	`string`	**Required**	Full legal name of the student (Max 200 chars).
`date_of_birth`	`date (YYYY-MM-DD)`	**Required**	e.g. `"2014-05-15"`
`academic_year`	`string / int`	**Required**	e.g. `"2026-2027"` or integer ID `1`
`class`	`string / int`	**Required**	e.g. `"Class 8"`, `"8"`, or integer ID `1`
`section`	`string / int`	**Required**	e.g. `"A"`, `"B"`, or integer ID `1`
`gender`	`string`	Optional	`"Male"`, `"Female"`, or `"Other"` (Default: Male)
`blood_group`	`string`	Optional	`"A+"`, `"A-"`, `"B+"`, `"B-"`, `"O+"`, `"O-"`, `"AB+"`, `"AB-"`, `"Unknown"`
`roll_number`	`string`	Optional	e.g. `"14"` (Max 20 chars)
`admission_number`	`string`	Optional	Auto-generated by backend if omitted (e.g. `"A-1021"`)
`admission_date`	`date`	Optional	Defaults to today (YYYY-MM-DD)
`status`	`string`	Optional	`"Active"`, `"Transferred"`, `"Graduated"`, `"Inactive"`
`bus_route_info`	`string`	Optional	e.g. `"Bus Route 4"`
`father_name`	`string`	Optional	Father's full name
`mother_name`	`string`	Optional	Mother's full name
`relation`	`string`	Optional	`"Father"`, `"Mother"`, or `"Guardian"`
`mobile_number`	`string`	Optional	Primary parent phone number (e.g. `"9840012345"`)
`alternate_mobile`	`string`	Optional	Secondary contact number
`parent_email`	`string`	Optional	Parent email address
`address`	`string`	Optional	Residential street address
`student_photo`	`file`	Optional	Student passport image (.jpg, .png)
`birth_certificate`	`file`	Optional	Birth certificate document (.pdf, .jpg)
`transfer_certificate`	`file`	Optional	Transfer certificate (.pdf, .jpg)
`aadhaar_card`	`file`	Optional	Government ID document (.pdf, .jpg)

COMPLETE RESPONSE (201 CREATED) — ALL 3 STEPS INCLUDED

```
{
  "id": 108,
  "admission_number": "A-1021",
  "full_name": "Rohan Verma",
```

```
  "date_of_birth": "2014-08-14",
  "gender": "Male",
  "blood_group": "O+",
  "academic_year": "2026-2027",
  "academic_year_name": "2026-2027",
  "class": "Class 8",
  "school_class": "Class 8",
  "class_name": "Class 8",
  "section": "A",
  "section_name": "A",
  "class_display": "Class 8 - A",
  "roll_number": "14",
  "admission_date": "2026-10-21",
  "status": "Active",
  "bus_route_info": "Bus Route 4",
  "father_name": "Suresh Verma",
  "mother_name": "Anita Verma",
  "relation": "Father",
  "mobile_number": "9840012345",
  "alternate_mobile": "9840098765",
  "parent_email": "suresh.verma@example.com",
  "email": "suresh.verma@example.com",
  "address": "12 Anna Salai, Chennai, TN",
  "student_photo": "http://127.0.0.1:8000/media/students/photos/rohan.jpg",
  "birth_certificate": "http://127.0.0.1:8000/media/students/documents/bc_108.pdf",
  "transfer_certificate": null,
  "aadhaar_card": "http://127.0.0.1:8000/media/students/documents/aadhaar_108.pdf"
}
```


GET PUT PATCH DELETE **/api/list-students/ | /api/view-student/<id>/ | /api/edit-student/<id>/ | /api/delete-student/<id>/**

Bearer Token Required

List Students: `GET /api/list-students/?search=Rohan&class=Class 8§ion=A&status=Active&page=1`
View Profile: `GET /api/view-student/108/` returns complete student dossier with academic records.
Edit Student: `PUT /api/edit-student/108/` or `PATCH /api/edit-student/108/` updates details and parents.
Delete Student: `DELETE /api/delete-student/108/` permanently deletes the student (204 No Content).

EDIT REQUEST EXAMPLE (PATCH /API/EDIT-STUDENT/108/)


```
{
  "roll_number": "15",
  "mobile_number": "9840099999",
  "status": "Active"
}
```


EDIT RESPONSE (200 OK)


```
{
  "id": 108,
  "full_name": "Rohan Verma",
  "roll_number": "15",
  "mobile_number": "9840099999",
  "status": "Active"
}
```

GET POST **/api/student-stats/ | /api/bulk-allocate-students/ | /api/export-students/**

Bearer Token Required

Student Stats: Returns enrollment KPI metrics, class-wise breakdown, and active tallies.
Bulk Allocate: Moves selected student IDs to a target section in bulk.
Export: Downloads CSV spreadsheet of student records.

GET /API/STUDENT-STATS/ RESPONSE (200 OK)


```
{
  "total_students": 450,
  "active_students": 420,
  "transferred_students": 12,
  "graduated_students": 15,
  "new_admissions_this_month": 28
}
```

6. Class & Section Allocation (Action-Based APIs)

Base: /api/

Manages physical classrooms, teacher assignments, and student seat limits (Capacity). Connects SchoolClass, Section, and AcademicYear models into assigned cohorts.

GET /api/list-class-sections/ or /api/class-sections/

Bearer Token Required | application/json

Retrieves allocated sections, assigned teachers, room numbers, and current strength vs capacity limits.

RESPONSE (200 OK)

```
[
  {
    "id": 1,
    "academic_year": 1,
    "academic_year_name": "2026-2027",
    "school_class": 1,
    "class_name": "Class 6",
    "section": 1,
    "section_name": "A",
    "display_name": "Class 6 - A",
    "room_number": "Room 101",
    "class_teacher_name": "Mrs. Anjali Rao",
    "current_strength": 38,
    "max_strength": 40
  }
]
```

POST /api/add-class-section/ or /api/class-sections/

Bearer Token Required | application/json

Configures a new class section allocation. Accepts string names or integer IDs for class, section, and academic year.

REQUEST BODY EXAMPLE

```
{
  "academic_year": "2026-2027",
  "school_class": "Class 7",
  "section": "A",
  "room_number": "Room 204",
  "class_teacher_name": "Mr. Vikram Sen",
  "max_strength": 45
}
```

RESPONSE (201 CREATED)

```
{
  "id": 5,
  "class_name": "Class 7",
  "section_name": "A",
  "display_name": "Class 7 - A",
  "current_strength": 0,
  "max_strength": 45
}
```

GET PUT DELETE **/api/view-class-section/<id>/ | /api/edit-class-section/<id>/ | /api/class-section-students/<id>/**

Bearer Token Required

View: `GET /api/view-class-section/1/` returns section details.
Edit: `PUT /api/edit-class-section/1/` updates teacher, room, or capacity.
Students in Section: `GET /api/class-section-students/1/` returns all enrolled students.
Delete: `DELETE /api/delete-class-section/5/` removes an empty allocation record.

STUDENTS IN SECTION (GET /API/CLASS-SECTION-STUDENTS/1/ RESPONSE)


```
[
  {
    "id": 101,
    "admission_number": "SAAPT-2025-001",
    "full_name": "Aarav Sharma",
    "roll_number": "12",
    "status": "Active"
  }
]
```

7. Authentication & Password Security

Base: /api/auth/

POST /api/auth/forgot-password/

No Auth Required | application/json

Forgot Password API (Step 1 - Ask Mail ID & Send OTP): Takes user's registered email address and sends a secure 6-digit OTP code (valid for 10 minutes).

REQUEST BODY

```
{
  "email": "user@saapt.com"
}
```

RESPONSE (200 OK)

```
{
  "message": "OTP has been sent to your registered email address.",
  "email": "user@saapt.com"
}
```

POST /api/auth/verify-otp/

No Auth Required | application/json

Verify OTP API (Step 2 - Verify Code): Validates the 6-digit OTP code entered by the user. If correct, returns a verification token for Step 3.

REQUEST BODY

```
{
  "email": "user@saapt.com",
  "otp": "582914"
}
```

RESPONSE (200 OK)

```
{
  "message": "OTP verified successfully. You can now enter your new password.",
  "email": "user@saapt.com",
  "reset_token": "u9mP0q-vA7Xz1bKr_8xL92aBcD4eFgHiJkLmNoPqRs"
}
```

POST /api/auth/reset-password/

No Auth Required | application/json

Reset Password API (Step 3 - Set New Password & Confirm): Takes the new password, confirmation password, and reset token to update the password.

REQUEST BODY

```
{
  "email": "user@saapt.com",
  "reset_token": "u9mP0q-vA7Xz1bKr_8xL92aBcD4eFgHiJkLmNoPqRs",
  "new_password": "NewSecurePass@2026",
  "confirm_password": "NewSecurePass@2026"
}
```

RESPONSE (200 OK)

```
{
  "message": "Password changed successfully. You can now log in with your new password."
}
```

POST /api/auth/login/ & /api/auth/register/

No Auth Required | application/json

Login API: Authenticates via username or email and returns JWT access & refresh tokens.
Register API: Staff account registration.

LOGIN REQUEST BODY

```
{
  "username_or_email": "user@saapt.com",
  "password": "NewSecurePass@2026"
}
```

LOGIN RESPONSE (200 OK)

```
{
  "message": "Login successful.",
  "user": {
    "id": 14,
    "username": "sarah_admin",
    "email": "user@saapt.com"
  },
  "tokens": {
    "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  }
}
```